

Bonds¹

Price (given yield) with one coupon period or less to redemption:

$$PRI = \left[\frac{RV + \frac{100 \times R}{M}}{1 + \left(\frac{DSR \times Y}{E} \right)} \right] - \left[\frac{A}{E} \times \frac{100 \times R}{M} \right]$$

where: PRI = dollar price per \$100 par value

RV = redemption value of the security per \$100 par value ($RV = 100$ except in those cases where call or put features must be considered)

R = annual interest rate (as a decimal; $CPN \div 100$)

M = number of coupon periods per year standard for the particular security involved (set to 1 or 2 in Bond worksheet)

DSR = number of days from settlement date to redemption date (maturity date, call date, put date, etc.)

E = number of days in coupon period in which the settlement date falls

Y = annual yield (as a decimal) on investment with security held to redemption ($YLD \div 100$)

A = number of days from beginning of coupon period to settlement date (accrued days)

Note: The first term computes present value of the redemption amount, including interest, based on the yield for the invested period. The second term computes the accrued interest agreed to be paid to the seller.

Yield (given price) with one coupon period or less to redemption:

$$Y = \left[\frac{\left(\frac{RV + R}{100 + M} \right) - \left(\frac{PRI}{100} + \left(\frac{A}{E} \times \frac{R}{M} \right) \right)}{\frac{PRI}{100} + \left(\frac{A}{E} \times \frac{R}{M} \right)} \right] \times \left[\frac{M \times E}{DSR} \right]$$

1. Source for bond formulas (except duration): Lynch, Jr., and Jan H. Mayle, *Standard Securities Calculation Methods*. New York: Securities Industry Association, 1986.

Price (given yield) with more than one coupon period to redemption:

$$PRI = \left[\frac{RV}{\left(1 + \frac{Y}{M}\right)^{N-1 + \frac{DSC}{E}}} \right] + \left[\sum_{k=1}^N \frac{100 \times \frac{R}{M}}{\left(1 + \frac{Y}{M}\right)^{k-1 + \frac{DSC}{E}}} \right] - \left[100 \times \frac{R}{M} \times \frac{A}{E} \right]$$

where: N = number of coupons payable between settlement date and redemption date (maturity date, call date, put date, etc.). (If this number contains a fraction, raise it to the next whole number; for example, 2.4 = 3)

DSC = number of days from settlement date to next coupon date

K = summation counter

Note: The first term computes present value of the redemption amount, not including interest. The second term computes the present values for all future coupon payments. The third term computes the accrued interest agreed to be paid to the seller.

Yield (given price) with more than one coupon period to redemption:

Yield is found through an iterative search process using the "Price with more than one coupon period to redemption" formula.

Accrued interest for securities with standard coupons or interest at maturity:

$$AI = PAR \times \frac{R}{M} \times \frac{A}{E}$$

where: AI = accrued interest

PAR = par value (principal amount to be paid at maturity)

Modified duration:²

$$\text{Modified Duration} = \frac{\text{Duration}}{1 + \frac{Y}{M}}$$

2. Source for duration: Strong, Robert A., *Portfolio Construction, Management, and Protection*, South-Western College Publishing, Cincinnati, Ohio, 2000.

where *Duration* is calculated using one of the following formulas used to calculate Macaulay duration:

- For a bond price with one coupon period or less to redemption:

$$Dur = \left(1 + \frac{Y}{M}\right) \cdot \frac{Dsr \times \left[\frac{Rv + \frac{100 \times R}{M}}{\left[1 + \left(\frac{Dsr \times Y}{E \times M}\right)\right]^2} \right]}{E \times M \times Pri}$$

- For a bond price with more than one coupon period to redemption:

$$Dur = \left(1 + \frac{Y}{M}\right) \cdot \frac{\left[\frac{Rv \times \left(N - 1 + \frac{Dsc}{E}\right)}{\left(1 + \frac{Y}{M}\right)^{N + \frac{Dsc}{E}}} \right] + \left[\sum_{k=1}^N \frac{100 \times \frac{R}{M} \times \left(k - 1 + \frac{Dsc}{E}\right)}{\left(1 + \frac{Y}{M}\right)^{k + \frac{Dsc}{E}}} \right]}{M \times Pri}$$

Note: Bond price formulas and notations are described in this section.

Depreciation

$RDV = CST - SAL$ - accumulated depreciation

Values for DEP , RDV , CST , and SAL are rounded to the number of decimals you choose to be displayed.

In the following formulas, $FSTYR = (13 - MO1) \div 12$.

Straight-line depreciation

$$\frac{CST - SAL}{LIF}$$

$$\text{First year: } \frac{CST - SAL}{LIF} \times FSTYR$$

$$\text{Last year or more: } DEP = RDV$$